

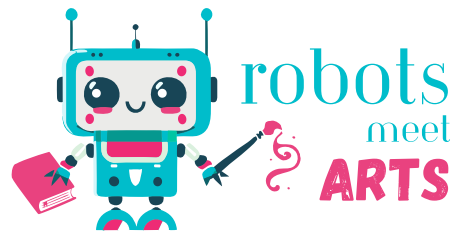


Co-funded by
the European Union

Project number: 2023-1-
FR01-KA220-SCH-
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GETTING STARTED WITH OCTO STUDIO

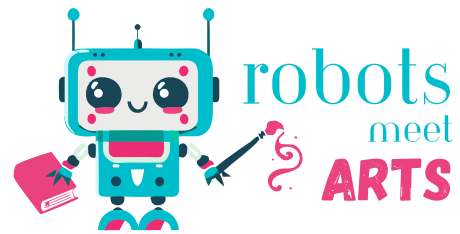




Why Choose OctoStudio Over other block-based programs, like Scratch?

OctoStudio offers a powerful mobile-first experience for creative coding. While programs like Scratch are well-known, OctoStudio stands out in several important ways:

- **Extensive learning resources:** Get started quickly with a rich library of video tutorials and demonstrations available on [OctoStudio's YouTube channel](#).
- **Designed for mobile, built for flexibility:** OctoStudio is designed specifically for mobile devices, enabling users to create projects anytime, anywhere—no computer required. Bring your projects to life with interactive features that respond to touch, shaking, tilting, and other mobile device sensors. Easily integrate photos, videos, and sounds from your surroundings into your projects, encouraging creativity that goes beyond the screen.
- **Free intuitive block-based coding:** With a user-friendly drag-and-drop interface, OctoStudio makes it easy for children aged 7 to 14 to learn coding concepts without worrying about syntax. It's free to use and, in many users' experience, more intuitive than Scratch.
- **Works Without Internet:** OctoStudio functions offline, making it an ideal tool for areas with limited or no internet access.
- **Multilingual and Accessible:** Available in over 20 languages and equipped with accessibility features like screen reader compatibility and image enlargement, OctoStudio is truly inclusive for all users.

**Subject:**

Coding

Content:

How to start plugged-in programming with Octo Studio?

**Computational practices:
choose what fits**

- **Decomposing**
- **Pattern recognition**
- **Abstraction**
- **Algorithms**
- **Problem solving**

Unplugged/**plugged:**

**Preparation time:**

100 min

**Age:**

8-12 years

**Duration:**

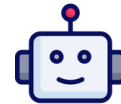
120 min

**Location:**

Classroom

**Group distribution:**

whole-class teaching and
pairteaching

**Difficulty****After this lesson, the children will be able to...**

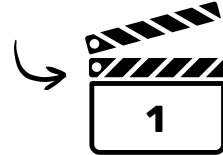
- **Create and customize a sprite using Octo Studio.**
- **Design a simple interactive program in Octo Studio that includes elements like movement, sound, text, and sensor input.**
- **Understand the meaning of the word 'pseudocode'.**
- **Work effectively in pairs by listening to each other's ideas and dividing tasks fairly.**
- **Communicate clearly with a partner to plan, build, and test a program.**
- **Practice problem-solving by debugging and improving their program.**
- **Show creativity in designing characters (sprites), backgrounds, and interactions.**



Materials needed:

For the teacher

- 1 x projector and cable to project from a tablet (app is online available on tablet)
- 1 x solution for challenge cards
- 1 x bell
- 1 x attachment with screenvideo's of the app. These symbols in this manual refer to a different video in the attachment.



For every pair

- 1 x charged tablet with Octo Studio app (free)
- 1 x challenge cards
- 1 x role cards (laminated)
- 1 x whiteboardmarker
- 1 x 'This is not a circle' worksheet
- 1 x colouring pencils

Note: All materials are ready to print and can be found at the end of this manual.

Helpful links:

- *Octo Studio reference guide:*
<https://octostudio.org/resources/Reference%20Guide.pdf>
- *Inspiration source for this lesson:*
<https://www.exploratorium.edu/tinkering/animate-shape>

Preparation and teacher notes

- Download the software on the tablets and select the correct language for your classroom.
- Prepare the materials as listed above.
- Optional: Divide the group into pairs in advance.
- Optional: Make the starter exercise to increase your background knowledge as a teacher.
- Optional: Read the Octo Studio reference guide with an explanation of every programming block



A snippet from the reference guide

Warm up: 'This is not a circle' (10' in pairs)

Have students sit in pairs. Distribute the worksheets and colouring pencils. Have the students colour in the circle with their favourite colour. *Then tell: Turn this circle into something else that is not a circle. Use your imagination.*

This is not a circle



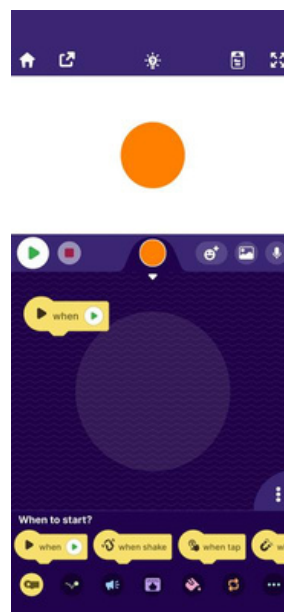
Devide the roles (5' in pairs)

Tell students the lesson objective: *to create a programme with our 'this is not a circle object' in the app Octo Studio.* Next, have them think about the two possible roles, since there is only 1 tablet. Eg: one kid to operate the tablet and one kid to say what the tablet operator should do. Let them write these roles on the cards with the whiteboardmarker and both choose one role.



Creating a sprite (5' whole class teaching)

Today we will be programming. To do this, we need something: a sprite or a background. For this project we are going to program a sprite. In class, create a new sprite in a new project by drawing the circle in the chosen colour (see video).



If there aren't enough tablets for each pair, consider using a microteaching approach: work with a small group while the remaining students engage with other classroom activities independently.

Getting to know the coding blocks (5' whole class teaching)

Ask students to press the Play button. Ask what happens. Nothing happens. This is because we haven't programmed anything yet. We are going to do that now. Tell the students that they will be using blocks to tell the sprite exactly what to do, step by step. We place the blocks from top to bottom, like building instructions. There are different types of blocks. Each kind has its own background colour. Ask for the meaning of each category:

Name	Description
 When to start?	Place one of these blocks on top of each script of code to tell it when to run.
 Motion	Tell your sprites how to move.
 Words and Sounds	Tell your sprite to play sounds, display text, and more.
 Scene	Make visual changes such as resizing your sprites and changing the backdrop.
 Colors and Light	Change the color of your sprite and other light settings.
 Control	Controls the flow of your code.
 More Blocks	Includes miscellaneous blocks such as variables, 'tilt', and custom blocks.

A snippet from the reference guide

Programming the challenge cards (45' in pairs)

Show the challenge cards to the students. Explain the meaning of the pseudo-code on these cards. *This is the step-by-step written explanation of what happens*

when you start the programme (press the play button). It is recommended to complete the first challenge together as a whole class. Students can then choose whether they want to complete Level 1 or Level 2. Level 2 is more difficult. In the top right corner of each challenge card, it shows which category of blocks they need to use. As a teacher, you may also choose to complete each exercise as a whole-class activity instead of letting students work in pairs. Before moving on to the next challenge, students must show their solution to the teacher.

Before they start challenge 8, the teacher should give them an additional instruction: video instruction 2. This way they can add a second sprite (their own design of the circle). In challenge 8 they program the circle and their own created sprite. There is a video of an example program in de PowerPoint.

Optional: Guess the circle game (20' whole class teaching)

Gather all children with tablets in the circle. Put a bell in the middle of the circle. Have students take turns showing their programme. As soon as another pupil recognises which new sprite they have made from the circle, they may ring the little bell.

Clean up (10')



Challenge 1

Level 1: The sprite jumps up 1 x.

Level 2: The sprite moves in the shape of a square.

Curious to explore more with OctoStudio? Jump into the next lesson and discover how to create your own animation!

List of annexes

AT_1: Role cards

AT_2: Worksheet

AT_3: Video 1

AT_4: Video 2

AT_5: Video 3

AT_6: Challenge cards

AT_7: Challenge cards solutions

AT_1: Role cards



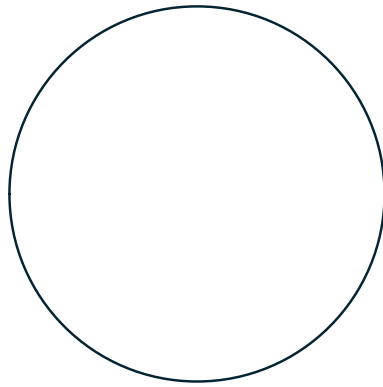
Role 1:



Role 2:

AT_2: Worksheet

This is not a circle

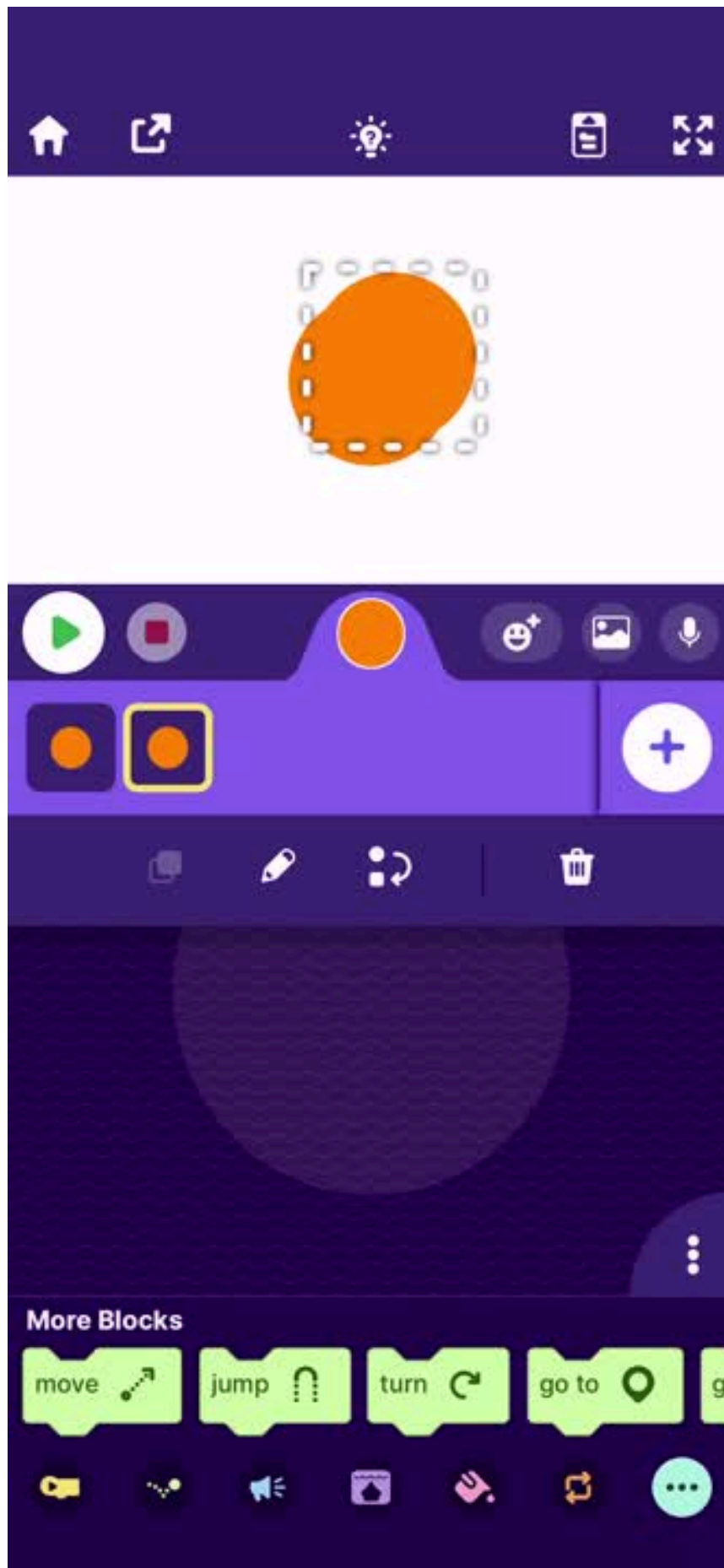


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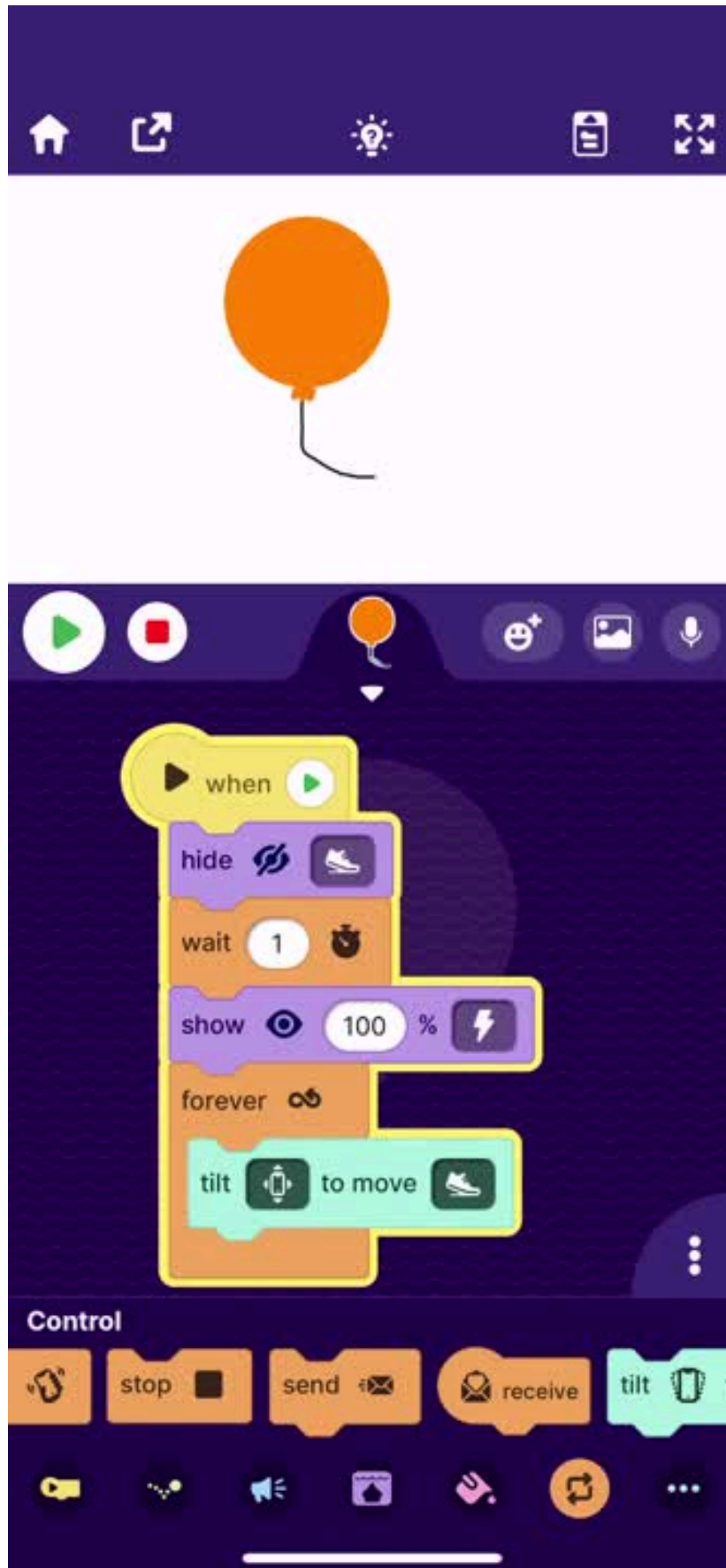
Creating a sprite



Edditing a sprite



Example programme



Challenge cards

Octo Studio



Challenge 1

Level 1: The sprite jumps up 1 x.

Level 2: The sprite moves in the shape of a square.

Challenge 2

Level 1: A high-pitched voice says ‘hello’.

Level 2: A sprite says ‘hello’ and you hear the words ‘hello’.

Challenge 3

Level 1: Hide the sprite fast.

Level 2: Resize the sprite.



Challenge 4

Level 1: Let the sprite glow.

Level 2: Change the colour of the sprite.



Challenge 5

Level 1: Hide the sprite for one second.

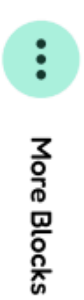
Level 2: The colour of the sprite changes forever.



Challenge 6

Level 1: When I shake the tablet, it plays a sound.

Level 2: When I touch the sprite, it changes colour.



Challenge 7

Level 1&2: The sprite follows my movements with the tablet.

Challenge 8

1. I see my original circle.
2. My original circle disappears after one second and I see my new sprite.
3. My new sprite gets bigger or I can move my new sprite with by moving the tablet.

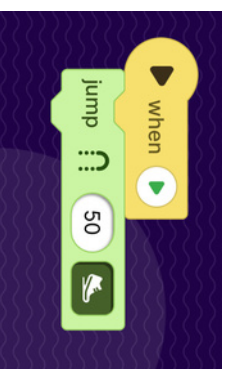
Challenge cards

Octo Studio

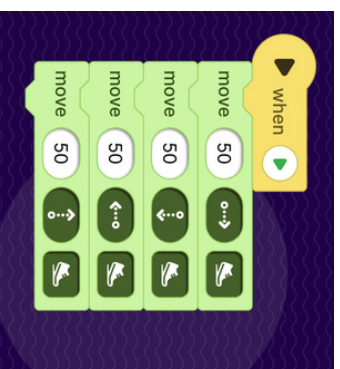


Challenge 1

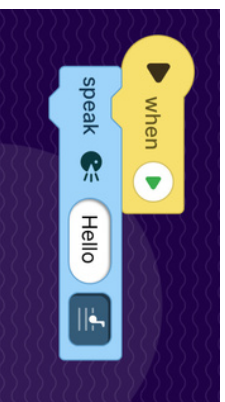
Level 1:



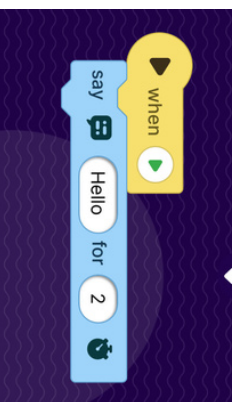
Level 2:



Challenge 2

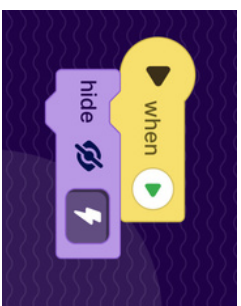


Level 1:

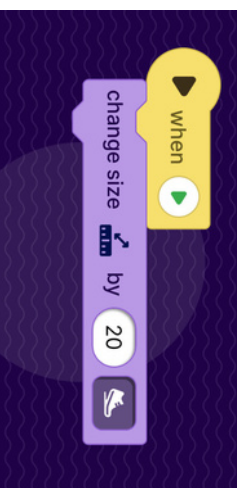


Level 2:

Challenge 3



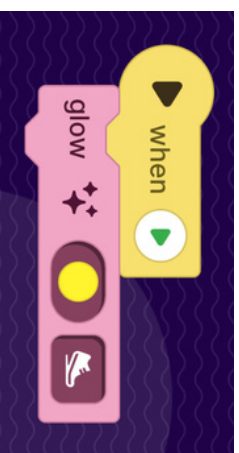
Level 1:



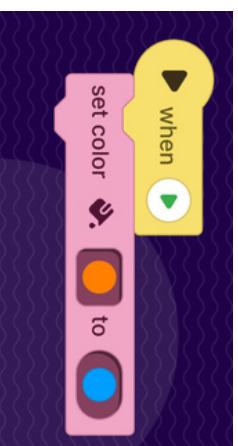
Level 2:

Challenge 4

Level 1:

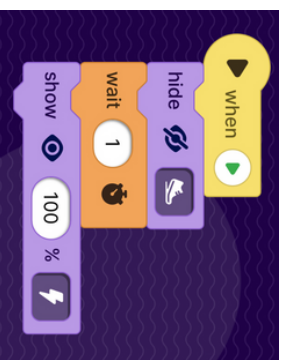


Level 2:

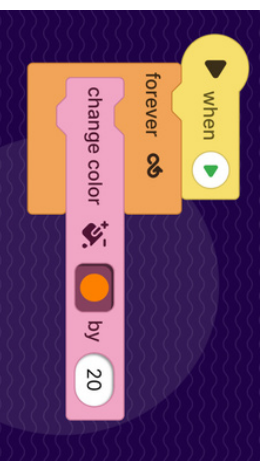


Challenge 5

Level 1:

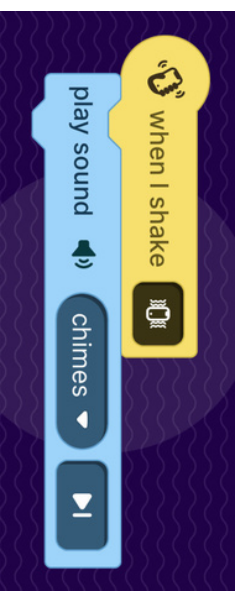


Level 2:

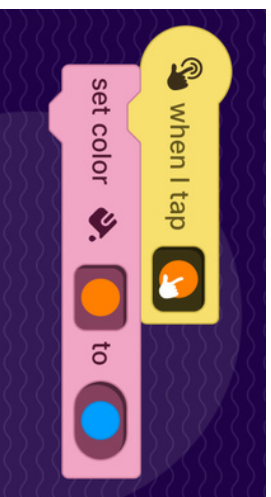




Challenge 6



Level 1:

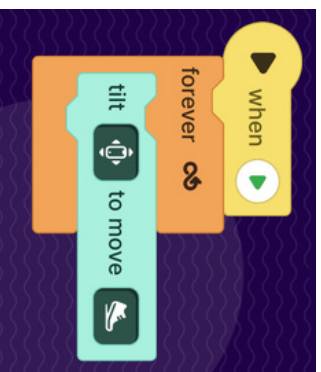


Level 2:

Challenge 7

More Blocks

Level 1&2:



Challenge 8

